

Software Engineering

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Cours Objectives

- This course introduces **methods** and **tools** to support the different phases of software development.
- The course is composed of 3 parts:
 - Software **modeling**: UML and architectural design patterns.
 - **Software testing** and **quality**: different test methods, quality criteria and metrics.
 - **Software development project** management: agile methods, DevOps.



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Course outline

- **General information on software engineering (1h)**
 - Definition of software engineering.
 - The phases of software development.
 - Software development cycle.
 - Approaches to software development.
 - Software.



Course outline

- **Software modeling with UML notation (7h)**
 - Introducing UML Notation
 - The class diagram
 - The Object and Sequence Diagram
 - The Transition State Diagram
 - The Use Case Diagram



Course outline

- **Software testing and quality (4h)**
 - Software testing
 - Software quality factors
 - Metrics
- **Design patterns (6h)**
 - Description of the Adapter, Observer, Editor, Visitor design patterns,...



Course outline

- **Software Development Project Management (12h)**
 - Agile methods.
 - DevOps.



Course organization

- **Courses**

- 20 slots of 1h30
- Lectures/Tutorials/Practical Exercises
- A final exam (modality and duration to be fixed)



Course schedule

- **Tutorials and Practical work**

- Tutorial / Lab session : UML modeling
- Tutorial / Lab session : design and test pattern
- Tutorial / Lab session : Agile methods
- Tutorial / Lab session : DevOps

Bibliographie

- **Software Engineering, 10th edition**

Pearson, 2021

Ian Sommerville

- *Introducing Software Engineering in as a whole (agile methods, project management, validation and testing, quality,....)*



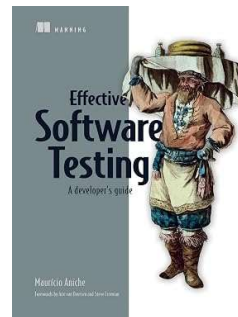
Bibliographie

- **Effective Software Testing: A Developer's Guide.**

Manning, 2022

Maurício Aniche

- *Methods Overview and testing techniques*



Bibliographie

- ***Head First Design Patterns, 2nd edition.***
Eric Freeman & Elisabeth Robson
O'Reilly, 2020

- *He explains the patterns from the start design issues rather than simply presenting a list of patterns.*

